GDD & MILESTONE PLANNING_

Turn your game idea into a clearly defined and realistically planned project.

EPITECH BERLIN / 2026–2027

CAPSTONE Project Sheet_

The Capstone Project Sheet is a guiding document designed to support you throughout the definition, planning and development of your Capstone Project. This version focuses on the Game Design Document (GDD) and on the definition and follow-up of your development milestones.

The GDD should provide a clear description of the game you intend to build. Your milestones then translate this design into concrete development objectives and expected results.

> **Important:** The more precise your milestone objectives are, the easier it is for you and the correctors to determine whether the planned work has been achieved. If an objective is too vague, its interpretation and evaluation will be left to the correctors' appreciation.

Submission Schedule_

	Students staying 1 semester	Students staying 2 semesters
GDD submission	12/10/2026	12/10/2026
Milestone 1	25/11/2026 — Only milestone	16/12/2026
Milestone 2	—	10/02/2027
Milestone 3	—	24/03/2027
Final Delivery	27/01/2027	26/05/2027

Two different development paths: Students staying for one semester have one milestone before their final delivery. Students staying for two semesters follow three development milestones before their final delivery.

Game Design Document (GDD)_

The GDD is the main design reference for your project. It should describe the game clearly enough for another person to understand what you are planning to build and what the intended player experience is.

Game Overview & Identity_

Before detailing gameplay systems, the GDD must first describe the project itself, so that correctors immediately understand what kind of game they are reading about and can put the design choices into context.

- Genre: The type of game you are making (e.g. platformer, FPS, puzzle, RPG, roguelike, narrative adventure).
- Setting & narrative context: The world, universe, story or theme in which the game takes place.
- Visual style / Art direction: The intended graphical style (e.g. pixel art, low-poly, realistic, cartoon, hand-drawn).
- Target platform(s): Where the game is intended to run (PC, console, mobile, VR, etc.).
- Target audience & tone: Who the game is made for and its overall tone (e.g. casual, hardcore, dark, humorous).

- Player fantasy / Core experience: What the player is meant to feel while playing.

Main Gameplay Mechanics_

Your GDD should define the main mechanics and systems that bring your game vision to life, including, where applicable:

- Gameplay Mechanics: What the player does and what systems support those actions.
- Movement: How the player moves and navigates the world (e.g. running, jumping, climbing, swimming, flying).
- Interaction: How the player interacts with objects, NPCs or the environment (e.g. item pickup, crafting, dialogue choices, switches).
- Combat, if applicable: Weapons, abilities, attack and defense systems, combos, magic, ranged mechanics, etc.
- Puzzle / Logic Systems, if applicable: The principles behind the puzzles (e.g. physics, time manipulation, pattern recognition).
- Special Mechanics / Core Innovation: The unique mechanic or system tied to the game's core concept.
- Progression & Rewards: How the player progresses, gains abilities or receives rewards.

What the GDD should make clear_

- What the player will actually do.
- How the main gameplay systems work.
- What content and systems are planned for the project.
- How the different systems interact with each other.
- What the intended game experience is.
- What the planned scope of the game is.

Be precise. The GDD should avoid descriptions that are so broad that the actual intended implementation cannot be understood. The goal is not to predict every line of code, but to describe the planned game and its systems clearly enough to guide development.

GDD Examples_

Too vague	Detailed enough
The player can fight enemies.	The player has a light attack, heavy attack and dodge. Light attacks can be chained up to three times; heavy attacks deal more damage but have a longer recovery. Dodging consumes stamina.
There will be several levels.	The final game will contain 5 playable levels. Each level is expected to take approximately 10–15 minutes to complete and introduces one main gameplay challenge.
There will be a progression system.	Players earn XP from completing encounters. Every 500 XP, the player gains a level and chooses one of three upgrades that affect movement, combat or health.

Milestones_

Milestones are development checkpoints. They are not simply dates on which the team reports what it has done. Each milestone should have clearly defined objectives and expected outputs that reflect the development stage of the project.

When planning a milestone, describe what you intend to complete during that period as precisely as possible. Avoid using objectives that cannot be meaningfully checked.

> **Milestones are deliveries:** Milestones must not be treated as informal check-ins or loosely-defined "big steps". Each milestone is a delivery in its own right, prepared and evaluated with the same seriousness as the final delivery.
>
> **Build required:** For every milestone and for the final delivery, a playable build of the project must be prepared and submitted. This build must reflect the objectives defined for that milestone/delivery and allow the correctors to actually test the work described.

How to define your milestone objectives_

1. Identify the concrete feature, system, content or improvement you intend to complete.
2. Describe what 'done' means for that objective.
3. Specify relevant quantities, limits, content, behaviours or expected results whenever possible.
4. Make sure the objective can realistically be completed within the milestone period.
5. Leave some time for unexpected problems, bugs, technical issues, integration work or changes in workload.

> **Plan for the unexpected:** It is strongly recommended to keep some time available in each milestone for unforeseen issues. A milestone plan that uses every available day for planned tasks leaves little room to react to problems discovered during development.

Good vs. vague milestone objectives

Too vague	Detailed enough
Improve the combat system.	Implement the light attack combo (3 hits), heavy attack, dodge and stamina system. Integrate them with the existing enemy prototype and ensure all four actions can be used in a complete combat encounter.
Add more enemies.	Implement 2 additional enemy types. Enemy A follows the player and performs a melee attack; Enemy B keeps distance and launches a projectile. Both must be playable in the existing test level.
Make the game more polished.	Replace placeholder player animations with the planned idle, walk, attack and damage animations; add the corresponding sound effects; and update the HUD to display health and stamina.
Create a playable level.	Create Level 2 with the planned three gameplay sections, one checkpoint, two enemy encounters and the final room. The level must be completable from start to finish without developer intervention.

Objectives are commitments, not permanent restrictions

Development plans can change. During development, you may discover that a feature is not working as expected, that the team's workload needs to be reorganized, that the project's vision has evolved, or that a technical constraint requires a different approach.

However, any modification to the objectives originally defined for a milestone or delivery must be requested to the pedagogical team, and accepted, at least 5 weeks before the deadline of the milestone or delivery concerned, before the new objectives are considered official.

This allows the team to adapt its project while keeping a clear and fair reference for milestone evaluation. Do not silently replace an objective after the milestone has started.

Students Staying 1 Semester

Students who stay for only one semester have a shorter development period. They have one milestone before their final delivery. The milestone must therefore be planned with particular attention to scope and feasibility.

25/11/2026 – Milestone 1

This is the only milestone for students staying one semester. It should demonstrate meaningful progress toward the final project and provide a clear picture of what has been implemented, what remains to be done, and whether the planned scope is realistic.

Your milestone plan should specify, at minimum:

- The concrete gameplay systems and features to be implemented.

- The content to be produced or integrated.
- The expected playable state of the game at the milestone.
- Any important technical or production objectives.
- The expected limitations or remaining work leading to the final delivery.

27/01/2027 – Final Delivery_

The final delivery is the final version of the project for students staying one semester. The milestone should therefore be used to establish realistic and clearly defined objectives that lead directly toward this delivery.

Students Staying 2 Semesters_

Students who stay for two semesters have three development milestones. The milestones should show a progressive development of the project, from proving the core concept to implementing the planned game and preparing a stable final version.

16/12/2026 – Milestone 1: Proof of Concept (PoC)_

Goal: The goal of the PoC is to demonstrate that the core idea of the game works and that the team has a realistic foundation for continuing development.

Expected outputs / evidence:

- Demonstrate the core gameplay loop or the most important part of it.
- Demonstrate the game's core or unique mechanic.
- Use a playable prototype or slice to prove the intended experience.
- Establish the necessary technical foundation for continued development.
- Confirm that the planned scope is realistically achievable.

Focus: The PoC does not need to be polished. Its purpose is to prove the concept, not to present a finished-looking game.

10/02/2027 – Milestone 2: Alpha Build_

Goal: The Alpha milestone focuses on implementing the planned core game and making it playable as a coherent experience.

Expected outputs / evidence:

- Implement the planned core gameplay systems.
- Integrate the main systems into the game flow.
- Have a playable version covering the intended structure of the game.
- Integrate a meaningful amount of the planned content.
- Identify and document known bugs, limitations and remaining work.
- Begin testing the overall balance and player experience.

Focus: The priority is functionality and completeness of the planned core experience rather than final polish.

24/03/2027 – Milestone 3: Beta Build_

Goal: The Beta milestone focuses on stabilization, integration and polish. The project should be approaching its final form.

Expected outputs / evidence:

- Integrate the planned art, audio, UI and other presentation elements.
- Fix major bugs and stabilize the game.
- Improve feedback and user experience.
- Optimize important performance issues.
- Ensure the planned game flow works from beginning to end.
- Identify and complete the remaining work required for final delivery.

Focus: The game should feel close to a final product. The remaining work should be clearly identified and manageable before the final delivery.

26/05/2027 – Final Delivery_

The final delivery is the final version of the Capstone Project for students staying two semesters. By this point, the project should have moved beyond the development objectives of the previous milestones and be ready for final presentation and evaluation.

Milestone Planning Template_

For each milestone, define your objectives using the following structure. The examples above illustrate the expected level of detail.

Section	What to provide
Objective	What specifically do you intend to achieve during this milestone?
Expected output	What concrete result, build, feature, content or document will demonstrate completion?
Definition of done	What must be true for the objective to be considered complete?
Scope / quantity	Where relevant, specify numbers, levels, enemies, mechanics, systems, assets, duration, etc.
Dependencies / risks	What could prevent the objective from being completed?
Buffer	What time have you kept available for unforeseen issues?

> **Remember:** A milestone objective should be specific enough that a person who was not involved in writing the plan can understand what the team intended to accomplish and can reasonably determine whether it was accomplished.

Post-Mortem_

In addition to the objectives themselves, a well-prepared post-mortem is required for each milestone and for the final delivery.

- What worked well, and why.
- What did not work as planned, and what was learned from it.
- How the objectives evolved compared to the initial plan, and why.
- What will be changed or improved for the next milestone/delivery.

> **Important:** Each post-mortem counts for 1 ECTS credit per milestone/delivery. A superficial or rushed post-mortem will be evaluated accordingly.

Final Checklist_

- The GDD clearly describes the planned game and its main gameplay systems.
- The milestone objectives are directly connected to the game described in the GDD.
- Objectives are as precise and measurable as reasonably possible.
- The expected output of each objective is clear.
- The team has avoided vague objectives such as 'improve', 'add more', 'make better' without further explanation.
- The planned workload is realistic for the time available.
- Some time has been reserved for unforeseen issues.
- Any future change to milestone objectives will be submitted for approval before being treated as an official change.
- A playable build is ready for each milestone and for the final delivery.
- A well-prepared post-mortem has been planned for each milestone and for the final delivery.